BTS-VI(SS)-01.19-0019 Reg. No. | | | | | | | | |

B

B.Tech. Degree VI Semester Special Supplementary Examination January 2019

ME 15-1603 OPERATIONS MANAGEMENT
(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Define crashing of networks.
(b) What is a dummy activity? Sketch on a network diagram.
(c) Give an example for moving average method of forecasting.
(d) What do you mean by product life cycle?
(e) What is a bin card?
(f) Differentiate between specialization and inter-changeability.
(g) Define the term aggregate planning.
(h) What is the function of Gautt Chart?
(i) Define the term "Fixed position layout".
(j) What is the concept of depreciation?

PART B

(4 × 10 = 40)

II. (a) What is the basic difference between PERT and CPM? (4)
(b) Briefly explain the importance of forecasting in operations management and how forecasting methods are classified. (6)

OR

III. The following table gives the activities in a construction project with the relevant information. (10)

Activity	Immediate Predecessor	Time (Months)		Direct cost (in lakhs)	
		Normal	Crash	Normal	Crash
A	-	4	3	60	90
B	-	6	4	150	250
C	-	2	1	38	60
D	A	5	3	150	250
E	C	2	2	100	100
F	A	7	5	115	175
G	D, B, E	4	2	100	240

Draw the network diagram, identify critical path, project duration before crashing and optimum project duration after crashing. Consider the indirect cost of the project per month as ₹60 lakhs.

IV. (a) What are the various functions of production planning and control? Explain each function with specific examples. (6)
(b) Differentiate between product life cycle and product consumption cycle. (4)

OR

(P.T.O.)

V. (a) Discuss the process stages involved in new product design. (5)

(b) What is EOQ? What are the costs involved in EOQ calculation? (5)

VI. (a) What is the role and need of aggregate planning in manufacturing sectors? (5)

(b) What are the steps involved in critical ratio method of loading and scheduling? (5)

OR

VII. (a) What are the objectives of sequencing and the assumptions made while solving sequencing problems? (5)

(b) A manufacturing company processes six different jobs on two machines A and B. Number of units of each job and its processing time on A and B are given in the following table. Find the optimum sequence, the total minimum elapsed time and idle time for each machine. (5)

Job Number	No. of units of each job	Processing Time (Hours)	
		Machine – A	Machine - B
1	3	5	8
2	4	16	7
3	2	6	11
4	5	3	5
5	2	9	7
6	3	6	14

VIII. (a) Discuss the factors that affect the selection of a plant layout with respect to different types of manufacturing operations. (5)

(b) Write notes on: (5)

(i) Systematic Layout Planning

(ii) Project Layout

OR

IX. (a) How are material handling equipments classified? Give examples. (5)

(b) What are the various principles of material handling? (5)

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